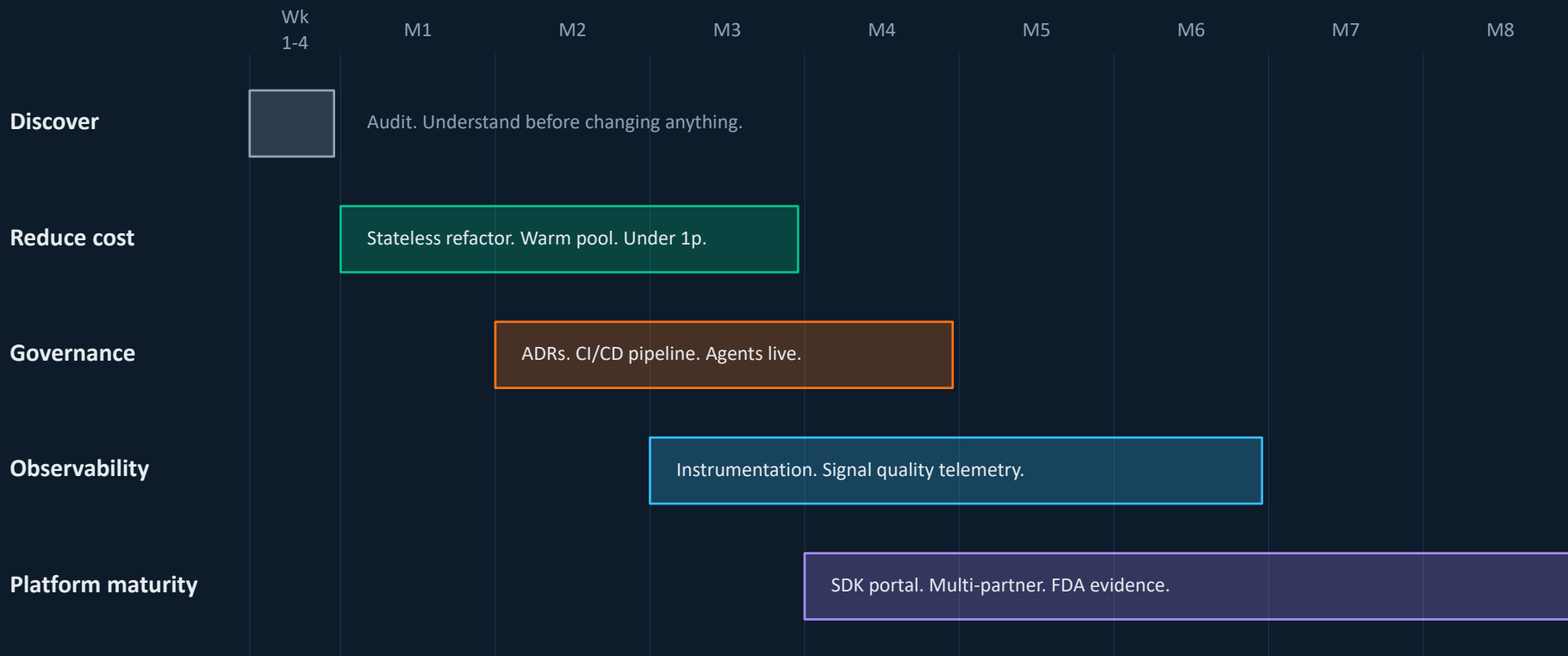


Incremental improvements without impacting production



Phases overlap. Sequencing reflects priority not exclusivity. MVP of each phase is brought forward where needed — e.g. instrumentation during the stateless refactor, not after.

Appendix — Platform Technology Stack



Current



Phase 1



Phase 2



Phase 3+

Cloud Infrastructure

Azure Container Apps	Container hosting, warm pool
AKS	Container orchestration at scale
Azure Service Bus	Async assessment queue
Azure Cache for Redis	Stateless session state
ACR	Immutable artefact registry
Terraform	Infrastructure as Code

Backend & Inference

Python / Flask	Current inference service
FastAPI	Target async inference service
Triton Inference Server	ML model serving at scale
MLflow	Experiment tracking, model registry

Signal Processing

POS + CHROM algorithms	rPPG signal extraction
SNR-weighted fusion	Best signal quality combination
Signal quality gate	Reject low-SNR before transmission

Mobile — iOS

Swift / SwiftUI	iOS SDK language and UI
Apple Vision Framework	Face landmark detection (native)
Metal	GPU compute for signal extraction
Fastlane	Build automation and release

Mobile — Android

Kotlin	Android SDK language
MediaPipe	Cross-platform face landmark tracking
Vulkan	GPU compute for signal extraction

Testing & Verification

pytest	Unit and integration tests
TestRail	Test management, IEC 62304 §5.7 artefact
Renovate / Dependabot	Automated dependency updates
Bland-Altman analysis	Clinical performance validation

Developer Experience

GitHub Pages → Backstage	Docs and internal developer portal
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CI/CD Pipeline

GitHub Actions	Pipeline execution, immutable logs
SonarCloud	Static analysis quality gate
Snyk	Dependency and container vulnerability scan
Syft	SBOM generation (FDA cybersecurity artefact)
Trivy	Container image scanning
AI Code Review Agent	Structured PR review

Observability — Cloud

Grafana	Metrics dashboards and alerting
Prometheus	Infrastructure and service metrics
Loki	Structured log aggregation
Sentry	Error tracking, release correlation
PagerDuty	On-call routing and incident management
Azure Monitor	Native monitoring (considered)

Observability — Mobile

Sentry Mobile SDK	Crash rate, ANR, B2B integration errors
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